

Performance and Resource Analysis of MLP Forward Pass on LabVIEW FPGA

Magellan Scholar
McNair Senior Fellowship

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Problem

- Neural networks are typically run on CPUs/GPUs, which are power-hungry and introduce latency unsuitable for real-time, embedded, or edge applications
- FPGAs offer parallel, low-latency, low-power computation — attractive for ML inference where speed and efficiency matter
- However, implementing a neural network on FPGA hardware isn't a direct port of a software model:
 - Math must be reworked into fixed-point arithmetic
 - Pipeline timing must be carefully managed
 - Overflow and precision loss must be avoided
 - All while meeting strict hardware timing constraints

Approach

- Target: NI cRIO-9033 Xilinx FPGA, developed in LabVIEW FPGA
- Built a matrix multiplication pipeline using an XNode block inside a Single-Cycle Timed Loop (SCTL)
- Extended this into a MLP with a 75-50-20-2 architecture, entirely in fixed-point arithmetic
- Used Flat Sequence Structure between SCTLs, and measured computation latency with gated Tick Count delta measurements

Acknowledgements

- This project is supported by McNair Summer Program through the Molinaroli College of Engineering and Computing at the University of South Carolina
- This project is based upon work supported by the National Science Foundation grant numbers CCF - 1956071, CCF-2234921, and CPS - 2237696. Any opinions, findings, conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation



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Motivation

- **Long-term goal:** full-scale 128-32-16-2 MLP for classification (% healthy vs. % unhealthy) on FPGA hardware
- Original 128-32-16-2 MLP ran at 0.609 ms/inference, which is too slow for real-time use
- Adopted a Xilinx block-based pipeline architecture to drastically cut latency within current hardware limits
- Current on-chip resource limits constrain the implementable size to 75-50-20-2

What is an MLP?

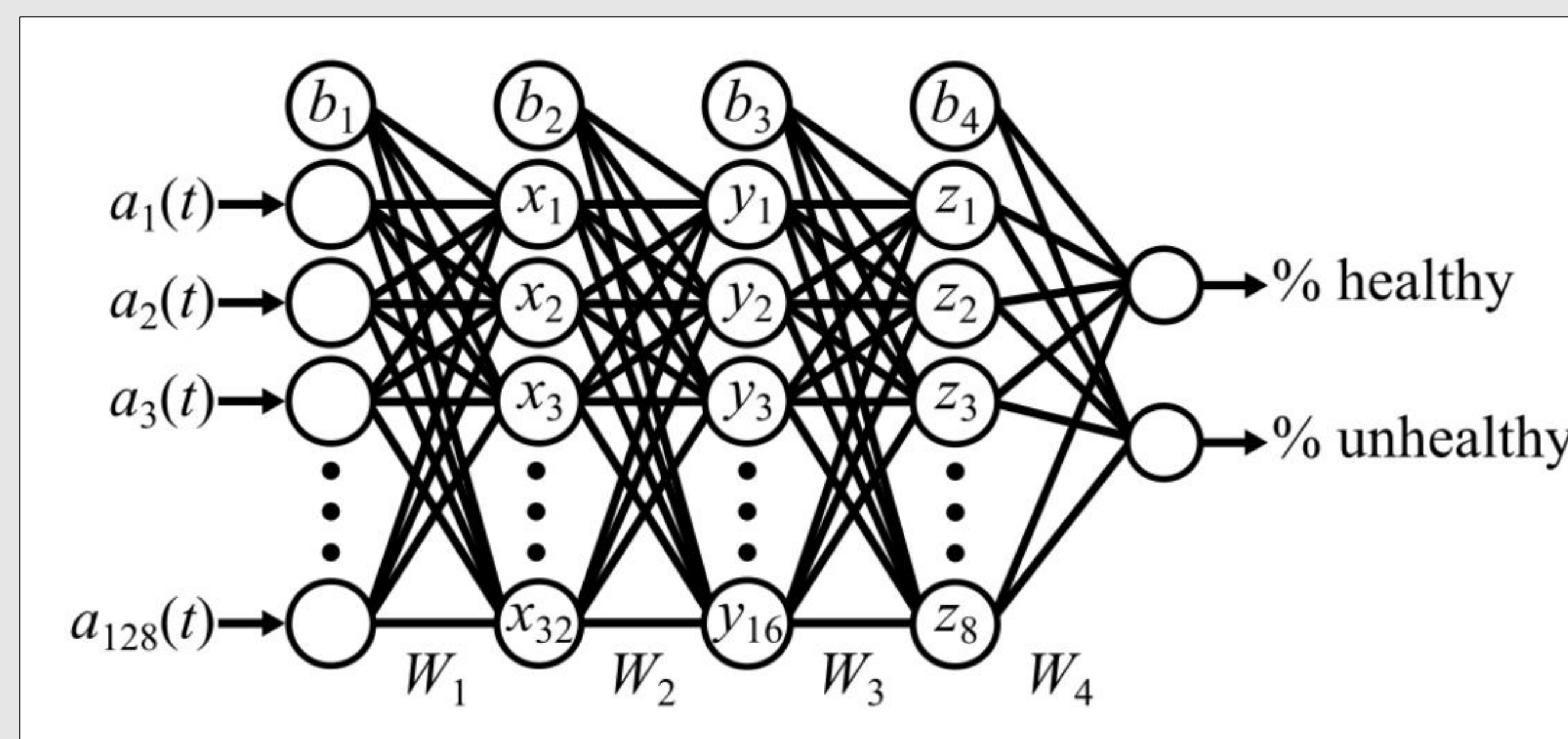


Figure 1: 128-32-16-8-2 MLP

- A multi-layer perceptron (MLP) is a type of forward propagation neural network made of stacked layers of nodes
- Data flows through a series of matrix multiplications followed by the addition of it bias values
- Mathematically, each layer computes:
 - $z = Wx + b$; where x is input vector, W is weight matrix and b is bias vector
- This structure lets the network learn complex, nonlinear mappings from inputs to outputs by adjusting weights during training

Resources

- ARTS-Lab. Ultra-High-Rate-State-Estimation GitHub, 2026, github.com/ARTS-Laboratory/Paper-2026-Ultra-High-rate-State-Estimation



Results

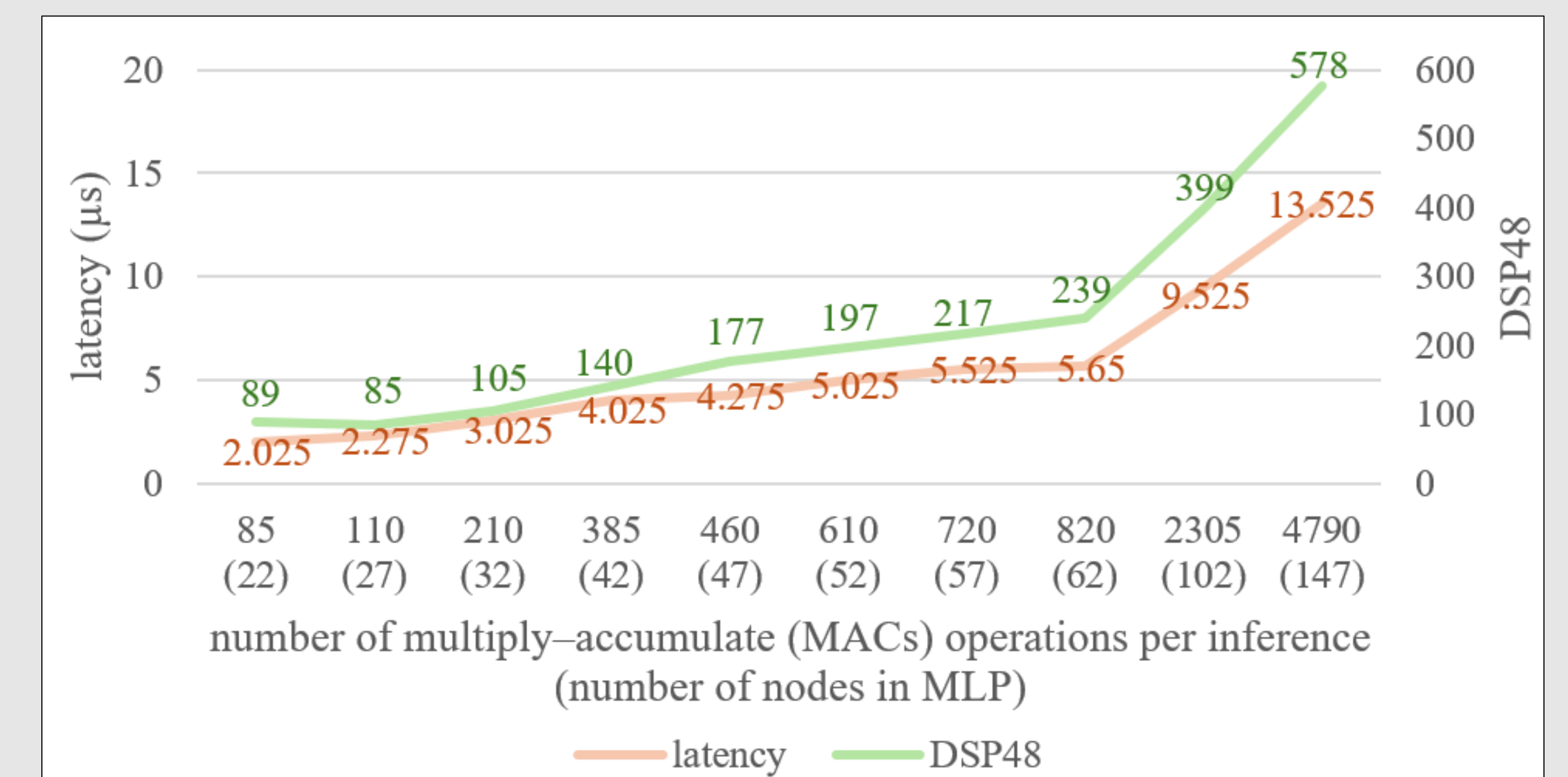


Figure 2: Latency and DSP48s for Different Sized MLPs

- Latency grows from 2.025 μs (85 MACs) to 13.525 μs (4790 MACs) — a ~6.7x increase for a ~56x increase in MACs
- Growth is roughly linear/gradual up to ~820 MACs, then steepens noticeably at the 2305 and 4790 MAC points — signaling diminishing efficiency as the network scales toward hardware limits

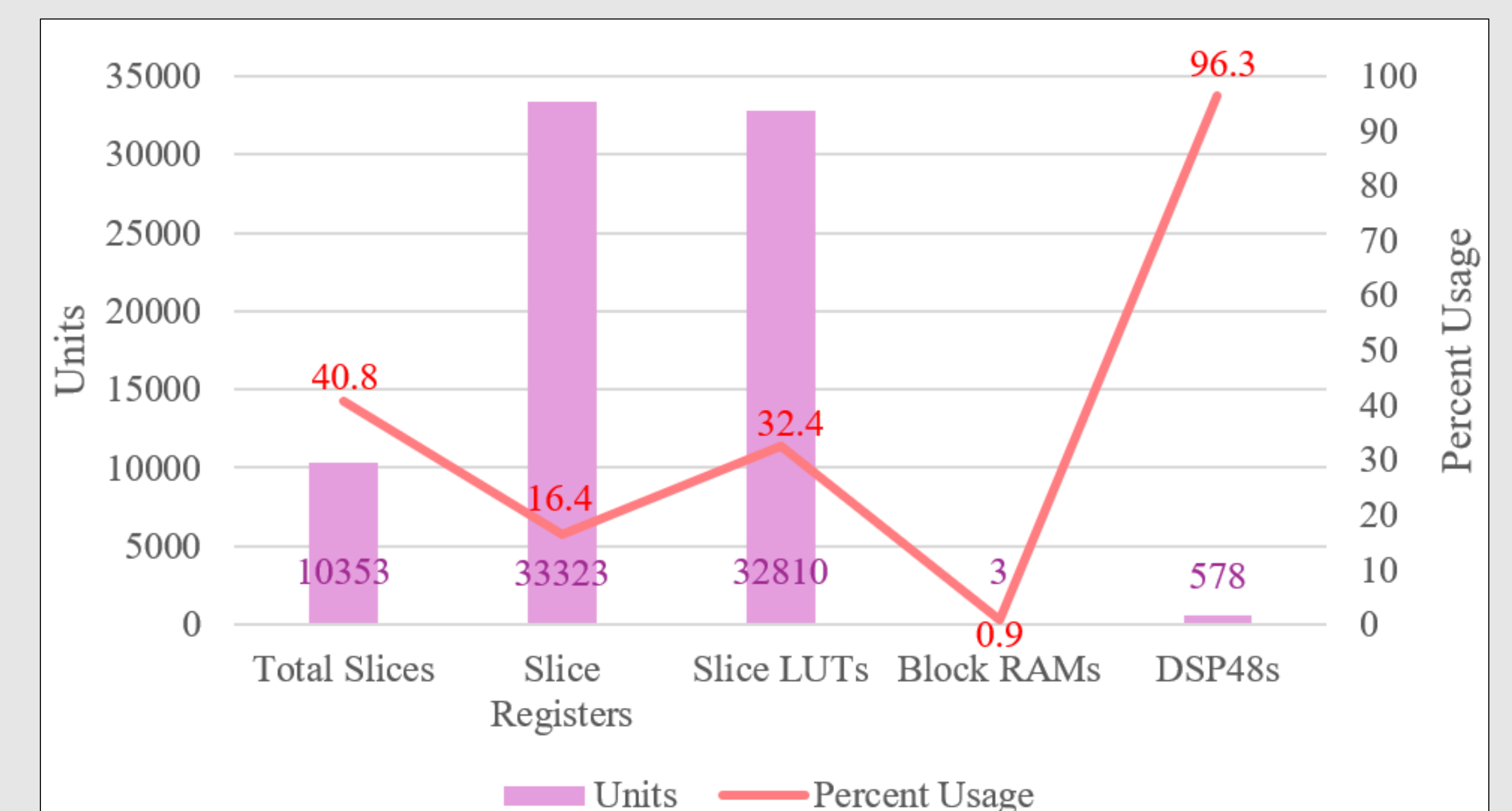


Figure 3: Resource Utilization for 4790 MACs, 147 Nodes

- DSP48s are the clear bottleneck, at 96.3% utilization (578 of 600 available), clearly far higher than any other resource